

CORRECTION OF CEREBROSPINAL FLUID PROTEIN IN INFANTS WITH TRAUMATIC LUMBAR PUNCTURES

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Abstract: In our multicenter cohort of infants ≤ 60 days of age, we identified 2646 infants with a traumatic lumbar puncture, of which 31 (1.2%) had bacterial meningitis. For every 1000 cerebrospinal fluid red blood cells/mm³, cerebrospinal (cerebrospinal fluid) protein increased 1.1 mg/dL (95% confidence interval: 1.0–1.2 mg/dL).

Key Words: bacterial meningitis, cerebrospinal fluid, protein, infants

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T.W.L. conceptualized and designed the study, acquired, analyzed and interpreted the data and drafted the initial manuscript. A.T.C. and S.B.F. conceptualized and designed the parent study, oversaw data acquisition and analysis and aided with drafting the manuscript. J.L.A., P.L.A., A.H.F., D.M.K., P.M., R.D.M., C.M.P. and A.D.T. acquired and interpreted data. L.E.N. conceptualized and designed the study, conceptualized and designed the parent study, oversaw data acquisition, analyzed and interpreted the data and drafted the initial manuscript. All authors critically revised the manuscript for intellectual content and approved the final manuscript as submitted.

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Febrile infants ≤ 60 days of age who present to the emergency department (ED) are frequently evaluated for bacterial meningitis with a lumbar puncture (LP).¹ Because cerebrospinal fluid (CSF) culture results require at least 24 hours to reliably exclude

bacterial growth, clinicians combine immediately available clinical and laboratory factors, including CSF protein, to assess risk of bacterial meningitis.² An elevated CSF protein has been associated with a higher risk of bacterial meningitis in children and may lead to empiric bacterial meningitis coverage.³ However, up to one-third of LPs in infants will be traumatic,^{4,5} increasing both CSF red blood cell (RBC) count and CSF protein, and limiting clinicians' ability to utilize CSF protein values in assessing an infant's risk of bacterial meningitis. Prior single-center studies have shown that the presence of RBCs in the CSF raises the CSF protein between 1.1 and 1.9 mg/dL for every additional 1000 CSF RBCs.^{6–10} The ability of CSF protein corrected for CSF RBCs to discriminate between infants with and without bacterial meningitis has not been examined.

We assembled a large, multicenter cohort of infants ≤ 60 days of age with traumatic LPs. We examined the relationship between CSF protein and CSF RBCs and assessed the ability of corrected and uncorrected CSF protein to identify infants with bacterial meningitis.

MATERIALS AND METHODS

We performed a planned secondary analysis of a retrospective cohort study of infants who presented to the ED of one of the 23 participating centers in the Pediatric Emergency Medicine Clinical Research Committee (PEM CRC) Herpes Simplex Virus (HSV) study (details of study protocol previously described).⁵ We limited our analysis to 20 sites which contributed CSF cell counts and protein. The study was approved by each of the participating institutional review boards.

We included infants ≤ 60 days of age presenting to a participating ED between January 1, 2005, and December 31, 2013, who had a CSF culture obtained within 24 hours of ED presentation and who had both CSF cell counts and chemistries available. We limited our analysis to those with a traumatic LP defined as a CSF RBC $\geq 10,000$ cells/mm³, a level of peripheral blood contamination of the CSF that makes the interpretation of CSF protein challenging.⁴ Infants with CSF RBCs $\geq 1,000,000$ cells/mm³ were excluded, as we could not confirm the presence of CSF in these samples. Infants could be included more than once if they had more than 1 eligible ED encounter.

Site investigators abstracted the following data elements from the electronic medical record: patient demographics, laboratory results (complete blood count, CSF cell counts and protein) and microbiologic results [blood, urine and CSF cultures and viral testing [HSV and enteroviral polymerase chain reaction (PCR) testing] when performed].

Our primary outcome was bacterial meningitis; defined as the growth of a bacterial pathogen in CSF culture.⁵ We classified the following pathogens as contaminants: coagulase-negative staphylococci, *Viridans streptococci*, *Propionibacterium acnes* and *Corynebacterium* species.

First, we explored the relationship between CSF protein and CSF RBCs using regression with CSF protein as the dependent variable and CSF RBCs as the independent variable. We decided a priori to assess the relationship as linear, based on prior evaluations,^{6,7} and to enhance clinical applicability, as nonlinear correction factors are difficult to apply.

Second, we calculated corrected CSF protein as follows: corrected CSF protein = CSF protein – (CSF RBC $\times$ correction factor), where the correction factor was determined using the beta coefficient from our regression. A normal CSF protein value was defined as ≤ 115 mg/dL for infants 0–28 days of age and CSF protein ≤ 89 mg/dL for infants 29–60 days of age.²

We calculated the following test characteristics for the corrected and uncorrected CSF protein as a predictor of bacterial

meningitis: sensitivity, specificity, negative predictive value, positive predictive value, positive likelihood ratio and negative likelihood ratio. Finally, we calculated the area under the curve from the receiver operating characteristic curves for both corrected and uncorrected CSF protein.

The analyses and figure for this paper were generated using SAS software version 9.4 (SAS Institute Inc., Cary, NC; 2015).

RESULTS

We identified 23,618 LPs, of which 19,819 (83.9%) had both CSF RBC counts and CSF protein available for analyses. Of these, 2659 (13.4%) were traumatic. We excluded 13 LPs with a CSF RBCs >1,000,000 cells/mm³, yielding a final sample of 2646 LPs (13.4% of eligible LPs). Median patient age was 25 days (interquartile range: 13–39 days), and 1442 (54.5%) were male. Thirty-one infants [1.2%; 95% confidence interval (CI): 0.8%–1.6%] had bacterial meningitis.

CSF protein increased 1.1 mg/dL (95% CI: 1.0–1.2 mg/dL) for every 1000 increase in the CSF RBC count (Fig. 1A). Using the 1.1 mg/dL correction factor, we found no difference between corrected and uncorrected CSF protein with regard to sensitivity, negative predictive value, positive predictive value, negative likelihood ratio or positive likelihood ratio for detection of bacterial meningitis (Fig. 1B). However, the specificity of corrected CSF protein was higher than the uncorrected CSF protein. While corrected CSF protein had slightly higher area under the curve for its

receiver operating characteristic curve for detection of bacterial meningitis compared with uncorrected CSF protein, discriminatory ability was low for both.

Of the 31 infants with bacterial meningitis, 4 infants (12.9%; 95% CI: 5.3%–29.0%) had normal corrected CSF protein but an elevated uncorrected CSF protein. All 4 of these misclassified infants were ≤28 days of age and were hospitalized on empiric antibiotics awaiting bacterial culture results. CSF cultures of these infants grew the following pathogenic organisms: *Staphylococcus aureus* (2), *Moraxella* species (1) and *Escherichia coli* (1).

One thousand six (38.0%) infants had HSV PCR testing of the CSF performed, and 4 out of 1006 (0.4%) were positive. Among these 4 infants, uncorrected CSF protein was elevated in 2 infants and corrected CSF protein was elevated in only 1 infant. Four hundred seventy-three (17.9%) infants had enteroviral PCR testing performed, and 66 out of 473 (14.0%) were positive. Among these 66 infants, uncorrected CSF protein was elevated in 48 infants and corrected CSF protein was elevated in 32 infants.

DISCUSSION

In our large, multicenter cohort of infants with traumatic LPs, CSF protein increased 1.1 mg/dL for every increase of 1000 CSF RBCs/mm³. Corrected CSF protein had a similar sensitivity but higher specificity when compared with uncorrected CSF protein. However, both corrected and uncorrected CSF protein alone had low discriminatory ability to identify infants with bacterial meningitis.

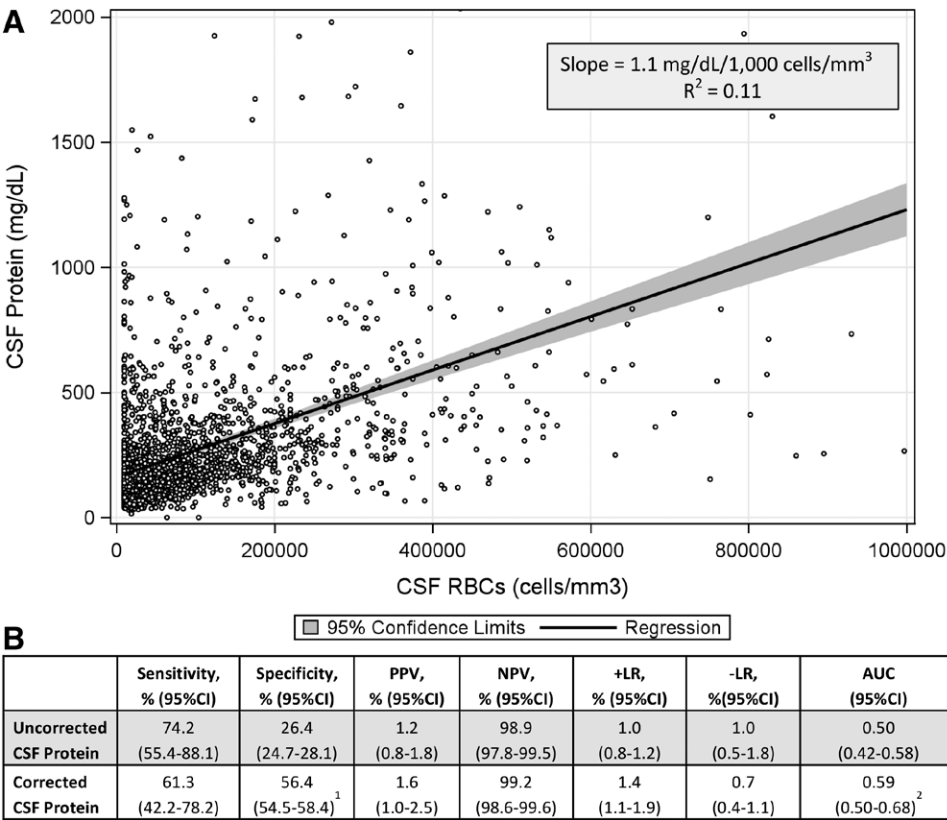


FIGURE 1. A: Plot of cerebrospinal fluid protein versus cerebrospinal fluid red blood cells in infants ≤60 days of age with a traumatic lumbar puncture. B: Test characteristics of uncorrected and corrected cerebrospinal fluid protein for the detection of bacterial meningitis in infants ≤60 days of age with a traumatic lumbar puncture. ¹Significantly different compared with uncorrected CSF protein at *P* < 0.001. ²Significantly different compared with uncorrected CSF protein at *P* = 0.00. AUC indicates area under curve for receiver operating characteristics curve; +LR, positive likelihood ratio; -LR, negative likelihood ratio; NPV, negative predictive value; PPV, positive predictive value.

Previous studies have evaluated the relationship between CSF protein and CSF RBCs in CSF.^{6–10} Our results are similar to a previous examination of 1354 children of all ages, which found a CSF protein-to-CSF RBC ratio of 1.1 mg/dL per 1000 CSF RBCs (95% CI: 0.9–1.1 mg/dL).⁶ Our results differed from another investigation of 1241 infants ≤56 days of age, which found a ratio of 1.9 mg/dL per 1000 CSF RBCs (95% CI: 1.7–2.1 mg/dL).⁷ Both studies examined this relationship in all CSF samples obtained, regardless of CSF RBC counts. In fact, the majority of children in both studies had nontraumatic LPs, where the CSF protein may not require correction.

Our findings add to the current understanding of the optimal interpretation of CSF protein in a young infant with traumatic LPs. First, our large, multicenter population increases the generalizability of our findings. Second, our analysis was limited to infants with traumatic LPs; thus, our results are applicable to a very large population for which CSF protein correction may be clinically indicated. Our study compared the predictive ability of corrected versus uncorrected CSF protein to determine the risk of bacterial meningitis. We found that neither corrected nor uncorrected CSF protein has adequate discriminative ability to be used alone for bacterial meningitis risk stratification. Therefore, clinicians should interpret CSF protein in combination with other clinical and laboratory values when assessing an individual infant's risk of bacterial meningitis. If clinicians incorporate CSF protein into this evaluation, corrected CSF protein offers higher specificity and slightly higher overall discriminatory ability compared with uncorrected CSF protein. However, this must be weighed against the risk of misclassifying a few additional infants with bacterial meningitis as having a normal corrected CSF protein.

Our study has limitations. First, despite a large sample of infants, bacterial meningitis was rare, and we may be underpowered to detect small difference in sensitivity of CSF protein. Second, we did not have data on clinical appearance, which may limit generalizability of our results. Third, not all infants underwent testing for HSV and enteroviral CSF infections. Therefore, we were unable to report the test characteristics of either uncorrected or corrected CSF protein for either enteroviral or HSV meningitis. Finally, as some study infants may have received antibiotic pretreatment before diagnostic LP rendering the CSF culture falsely negative, we may have misclassified infants as not having bacterial meningitis. However, few febrile infants are pretreated with antibiotics prior to diagnostic evaluation.¹¹ Therefore, we would caution against application of our results to infants with antibiotic pretreatment before LP.

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VARICELLA ZOSTER VIRUS-ASSOCIATED NECROTIZING RETINITIS AFTER CHICKENPOX IN A 10-YEAR-OLD FEMALE

A CASE REPORT

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Abstract: A necrotizing retinitis in children is a rare but vision-threatening ocular complication of chickenpox. We report a 10-year-old girl who developed chickenpox 1 month before presenting with panuveitis and necrotizing retinitis. After prompt antiviral treatment, her inflammatory signs were resolved. Early detection and treatment of varicella zoster-associated necrotizing retinitis after chickenpox can achieve good visual outcome.

Key Words: acute retinal necrosis, chickenpox, necrotizing retinitis, varicella zoster virus

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